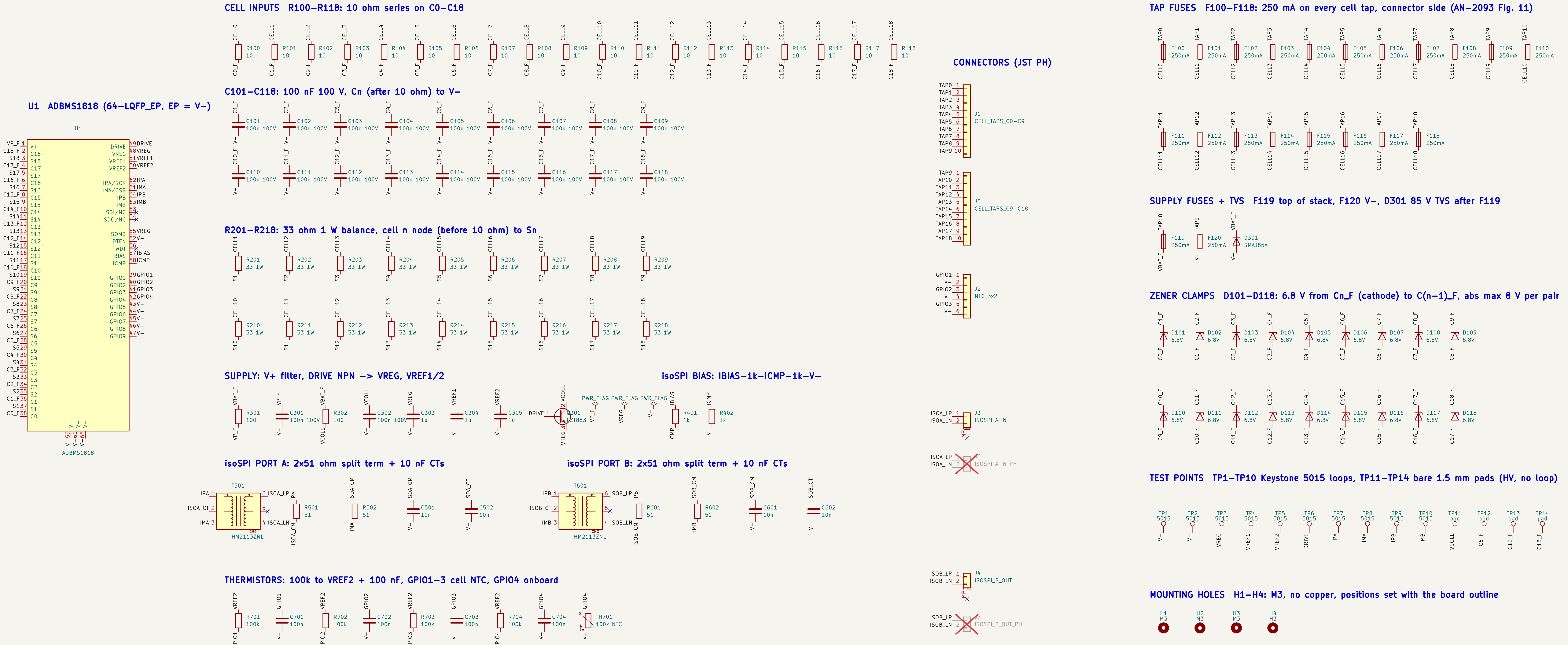
Wired with net labels: labels with the same name are one net. TAPn = connector side of fuse F1nn, CELLn = after it. V− (chip ground) has its own fuse F120.
ADBMS1818 Figure 112, isoSPI filtering per Fig. 104(B). See ../docs/slave−18s−design−basis.md.


CELL INPUTS R100–R118: 10 ohm series on C0–C18
C101–C118: 100 nF 100 V, Cn (after 10 ohm) to V−
R201–R218: 33 ohm 1 W balance, cell n node (before 10 ohm) to Sn
SUPPLY: V+ filter, DRIVE NPN −> VREG, VREF1/2
isoSPI BIAS: IBIAS–1k–ICMP–1k–V−
isoSPI PORT A: 2x51 ohm split term + 10 nF CTs
isoSPI PORT B: 2x51 ohm split term + 10 nF CTs
THERMISTORS: 100k to VREF2 + 100 nF, GPIO1–3 cell NTC, GPIO4 onboard
U1 ADBMS1818 (64–LQFP_EP, EP = V−)
U1
ADBMS1818
CONNECTORS (JST PH)
J1 CELL_TAPS_C0–C9
J5 CELL_TAPS_C9–C18
J2 NTC_3x2
J3 ISOSPI_A_IN
ISOSPI_A_IN_PH
J4 ISOSPI_B_OUT
ISOSPI_B_OUT_PH
TAP FUSES F100–F118: 250 mA on every cell tap, connector side (AN–2093 Fig. 11)
SUPPLY FUSES + TVS F119 top of stack, F120 V−, D301 85 V TVS after F119
F119 250mA
F120 250mA
D301 SMAJ85A
ZENER CLAMPS D101–D118: 6.8 V from Cn_F (cathode) to C(n−1)_F, abs max 8 V per pair
TEST POINTS TP1–TP10 Keystone 5015 loops, TP11–TP14 bare 1.5 mm pads (HV, no loop)
MOUNTING HOLES H1–H4: M3, no copper, positions set with the board outline
T501 HM2113ZNL
T601 HM2113ZNL
Q301 FZT853
TH701 100k NTC
PWR_FLAG

Not for manufacture
Baseline: ADBMS1818 datasheet Figure 112 typical application

Sheet: /
File: adbms1818−slave−18s.kicad_sch

Title: ADBMS1818 18S slave module

Size: User | Date: 2026−11−06 | **Rev: 0−draft**
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